

DEPARTMENT OF CIVIL ENGINEERING

CE331 PROJECT PRESENTATION

SURVEYING AND MAPPING

WESTERN LAB EXTENSION AREA

TUESDAY GRP-6

MEMBERS:

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INSTRUCTOR:

DR. SALIL GOEL

ACKNOWLEDGEMENT

We want to give our heartfelt thanks to our instructor, **Dr. Salil Goel**. His guidance and expertise were invaluable, but it was his support that truly enriched this learning experience for us.

We also owe a special thank you to our tutor, Mr. Praveen, his incredible patience and dedication were crucial. He was always willing to go the extra mile to help us, and that made a lasting impact on our progress.

Finally, we couldn't have finished this without the amazing **lab staff, our TAs, and our fellow group members**. Their commitment and assistance were vital, and we are deeply grateful for all their efforts.

INTRODUCTION

This project focuses on surveying and mapping of the Western Lab Extension area using modern surveying instruments such as Total Station, Auto-Level, Prism (Retroreflector), and GNSS.

The primary goal is to obtain accurate measurements of distances, angles, and elevations. These measurements will be utilized to generate a detailed map representing the site’s layout, boundaries, and major ground features with high precision.

OBJECTIVE

- Create a detailed topographic map with key site features.
- Establish accurate control points and compute global coordinates using GNSS.
- Perform precise data collection, processing, and error adjustments.
- Develop hands-on skills with Total Station, Auto-level, and QGIS.
- Create a reliable spatial reference for future use.

METHODOLOGY

Week 1 (Reconnaissance Survey): Established a four-point closed traverse by performing a reconnaissance survey of the assigned area. Each control point was marked with paint for easy future reference, enabling efficient planning.

Week 2 (Levelling): Conducted leveling across the site using an Auto-level. The GIS lab benchmark was used as a reference to ensure accurate elevation measurements.

Week 3 (Traversing & Angle Measurement): Used a Total Station to measure the distances and interior angles between control stations, formally establishing the closed traverse. Local coordinates were then calculated and adjusted using Bowditch's Rule for accuracy. **(Coordinates):** One of the control point was taken as the reference and the coordinates were assumed (1000,1000) . The point served as a reference to transform all control points from local to global coordinates using an Affine Transformation.

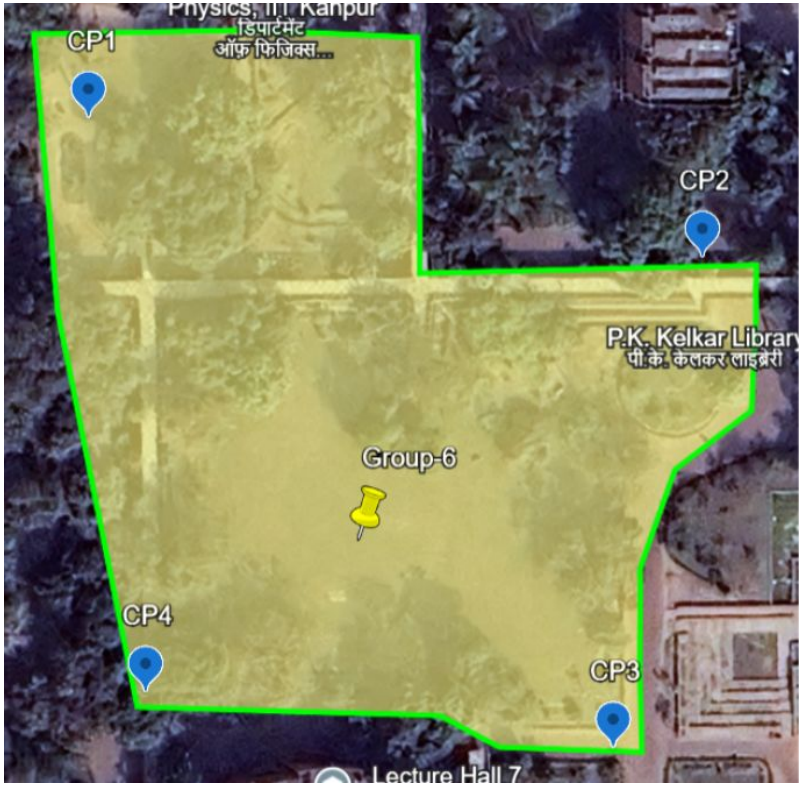
Week 4 & 5 (Feature Mapping): Mapped all significant site features, including trees, buildings, skywalks, manholes, and roads, using a Total Station. During this process, contour points were also recorded to create a detailed representation of the site's topography.

Week 6 (Mapping): Finalized the map by exporting the Total Station data to a CSV file and importing it into QGIS. Colors and symbols were assigned to all features based on Survey of India standards, and the map was exported for final presentation.

RECONNAISSANCE - WEEK 1

The objective of our initial site visit was to plan and optimize the main survey.

- **Assessed Site Conditions:** Conducted a preliminary inspection of the area.
- **Identified Key Elements:** Located key features (e.g., manholes, skywalk) and potential obstacles.
- **Established Control:** Selected optimal locations for equipment and set four control points (CP1-CP4) to define the survey traverse.
- **Ensured Accuracy:** This planning phase was vital to streamline the project, align our plan with site-specific factors, and improve the accuracy of all future measurements.



LEVELING - WEEK 2

Control Point	Adjusted RL
CP1	128.045
CP2	127.993
CP3	127.963
CP4	127.938

- Carried out leveling operations in the allotted area using an Auto-level instrument, taking the GIS lab benchmark as reference to establish precise elevation values.
- Misclosure = 2 mm
- Quality of work is Precise

KEY ASPECTS OF CONTROL POINT SELECTION

Stability & Accessibility: Control points are selected on stable, undisturbed, and easily accessible locations for long-term reliability.

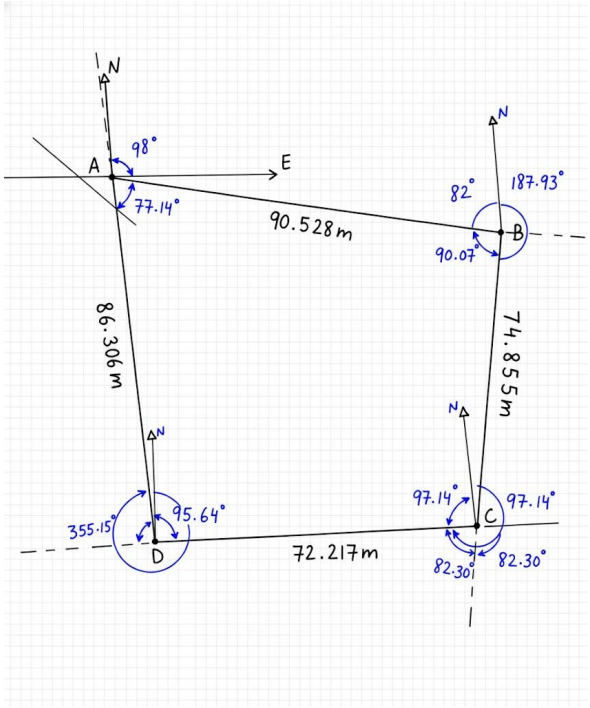
Visibility & Line of Sight: Clear visibility and unobstructed line of sight to other points ensure precise angle and distance measurement.

Interconnectivity: Points should be arranged to maintain strong geometric linkage, forming a reliable control network.

Spacing & Accuracy: Points are spaced based on project scale and required accuracy for effective data collection and mapping.

TRAVERSING - WEEK 3

POINTS	EASTING	NORTHING	ELEVATION
A	1000	1000	128.045
B	1089.699	987.3298273	127.993
C	1079.415	913.1318395	127.963
D	1007.247	914.0708164	127.938



- Total Station was used to measure side lengths and interior angles between control points in the closed traverse.
- Determined local coordinates for each control point.
- Adjusted coordinates to correct minor errors and improve accuracy.
- **Angle Misclosure** = 0°0'25"
- **Quality of Work** is between 1st order and 2nd order for Angle

MAP





SCALE 1:700

0 25 50 m

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MAP LEGEND

- Stairs
- Ramp
- Manholes
- Control Points
- Water Pump
- Water tank
- Light Pole
- Trees
- Skywalk
- Road
- Counters
- Ground

CONCLUSION

Project Learnings & Outcomes

- **Mastered Surveying Concepts:** Explored the core methodologies of geospatial positioning and modern mapping techniques.
- **Gained Field Expertise:** Acquired practical experience using sophisticated equipment, including Total Stations, Auto Levels, and prisms.
- **Established a Control Network:** Gained expertise in establishing and adjusting control points to create a precise survey framework.
- **Processed & Visualized Data:** Used QGIS to process all field data, transforming raw measurements into an accurate and detailed final map.

THANK YOU